ALIGNING VBENCH SCORES WITH THEIR INTENDED TARGETS: TOWARD FAITHFUL VIDEO EVALUATION

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ABSTRACT

Fine-grained video benchmarks rely on scoring implementations that faithfully measure their stated properties. We audit nine VBench dimensions and identify three recurring forms of target–evidence misalignment: Target Substitution, Nuisance Entanglement, and Evidence Collapse. Controlled counterfactual tests examine whether scores respond to changes in the target property and remain stable under irrelevant variation. We develop targeted repairs through semantic target parsing, region isolation, a learned motion readout, and preservation of discriminative information. The audit exposes unintended responses to synonymous wording, object and attribute bindings, background variation, local texture jitter, and spatial direction. It also reveals the limits of text-side repair: recovering ordered relation triples cannot restore direction already discarded by an unsigned geometric backend. This framework connects implementation-level diagnoses to explicit behavioral tests and clarifies which stage of a scoring pipeline requires repair.

***Index Terms*—** video generation evaluation, VBench, counterfactual audit, measurement validity, semantic alignment

1. INTRODUCTION

Recent advances in video generation have increased the need for reliable evaluation of semantic and temporal capabilities. FETV [1] and T2V-CompBench [2] examine prompt categories and composition, while VBench [3], VBench++ [4], and EvalCrafter [5] provide multidimensional evaluation. VBench-2.0 [6] extends this scope to intrinsic faithfulness, and VideoScore [7] learns from human feedback. Together, these efforts make dimension-level scores central to characterizing model strengths and weaknesses.

However, interpreting these scores requires that **each implementation measures its stated property** [8]. Our audit of VBench examines this premise through controlled input changes. In Scene, replacing "ocean" with "sea" can substantially lower the score for an unchanged video. Conversely, mirroring objects reverses "A is left of B" into "A is right of B," yet Spatial Relationship can assign identical scores through unsigned geometry. As illustrated in Fig. 1, these complementary failures penalize target-preserving changes and overlook violations of the target relation.

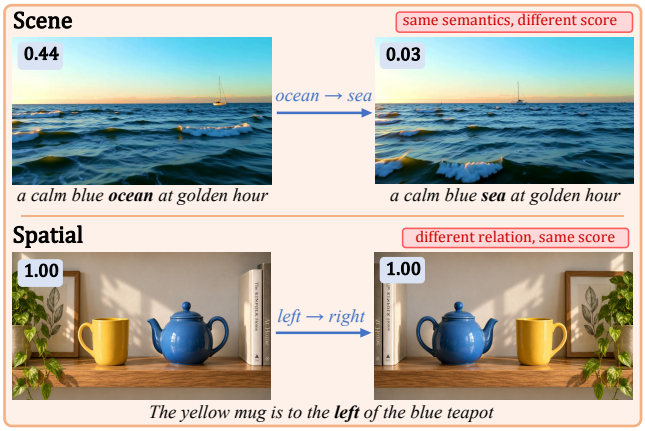


Fig. 1. VBench failures: scene synonyms and spatial reversals.

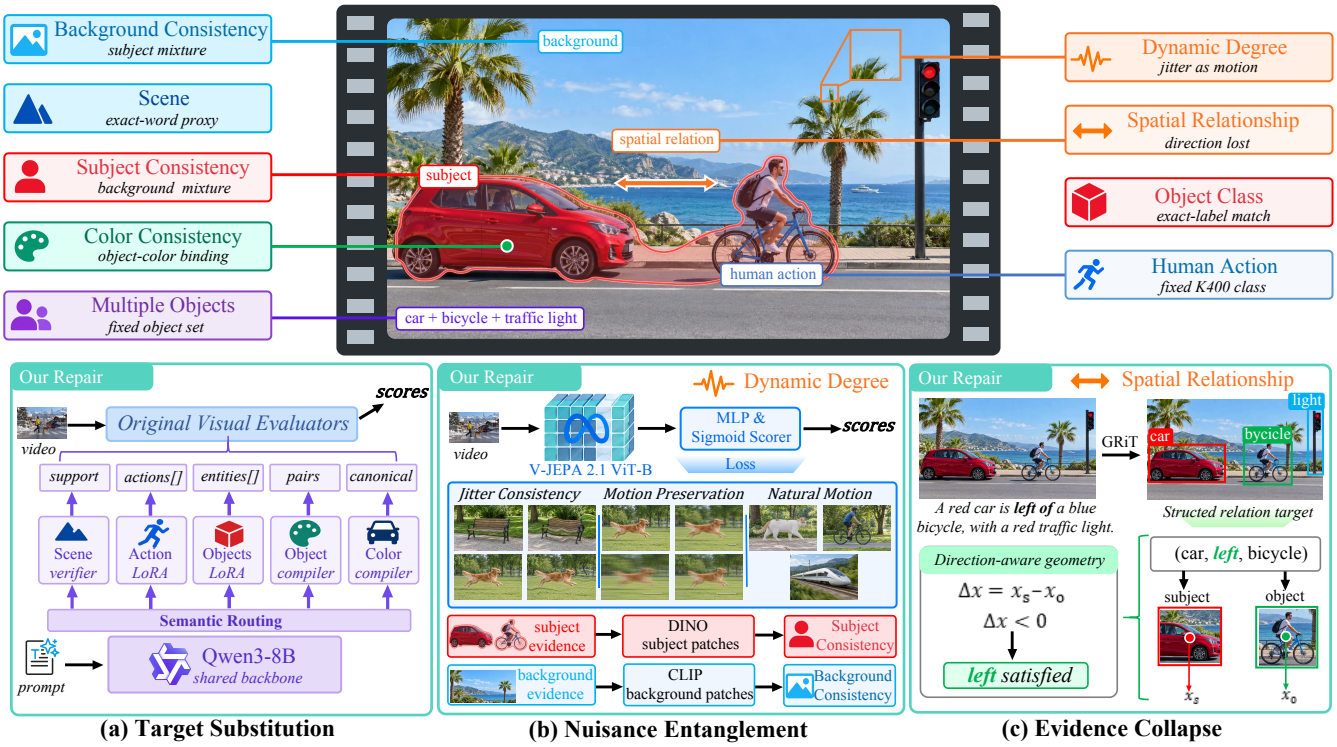
We characterize these failures as **target–evidence misalignment**: a mismatch between a dimension's intended property and what its scoring pipeline actually measures through evidence representation, selection, and aggregation. We identify three recurring mechanisms. **Target Substitution** replaces the intended target with an inequivalent proxy; **Nuisance Entanglement** mixes relevant and irrelevant evidence; **Evidence Collapse** discards discriminative information. These mechanisms can coexist within a dimension: spatial scoring substitutes unsigned separation for a directed relation while discarding object roles and direction information.

We diagnose these mismatches using controlled counterfactual audits [9, 10] that jointly test sensitivity to target changes and invariance to target-preserving perturbations. By changing one factor while fixing other content and scoring conditions as far as possible, we connect observed failures to specific pipeline stages, then assess the benefits and boundaries of repairs at those stages.

Our contributions are:

1. We audit nine VBench dimensions and identify three recurring target–evidence misalignment mechanisms: Target Substitution, Nuisance Entanglement, and Evidence Collapse.
2. We introduce controlled counterfactual audits that jointly test sensitivity to target changes and invariance to target-preserving perturbations, enabling stage-wise diagnosis of scoring failures.
3. We develop targeted repairs for semantic parsing, evidence isolation, motion scoring, and aggregation, substantially improving alignment.

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(a) Target Substitution (b) Nuisance Entanglement (c) Evidence Collapse

Fig. 2. Overview of target–evidence misalignment and repair strategies

2. ANALYSIS AND METHODOLOGY

We compare each target with its scoring evidence and aggregation, then repair unintended sensitivity or invariance (Fig. 2).

2.1. Target Substitution

Target Substitution replaces semantics with inequivalent proxies.

2.1.1. Scene and Human Action

Original scoring. Scene tests whether target terms occur in sampled frame captions from Tag2Text [11]. Human Action checks whether the target action occurs among UMT's [12] top-five Kinetics-400 (K400) [13] predictions.
Mechanism. Word or fixed-class matching substitutes for semantic agreement: equivalent descriptions can score differently, and recognized actions are restricted to K400.
Repair. Keeping the visual evidence fixed, a Qwen3-8B [14] Scene Verifier labels prompt–caption pairs as *supported*, *contradicted*, or *insufficient*. An Action Adapter parses actions and canonicalizes them to K400 for comparison with the original UMT predictions. A deterministic interface handles declared K400 synonym pairs.

2.1.2. Multiple Objects and Color

Original scoring. Multiple Objects checks joint object presence by splitting a preset pair at `and` and exactly matching both GRiT [15] labels. Color deletes articles and color words from the prompt to form an object query; after exact label matching, it checks color substrings in descriptions.
Mechanism. Fixed strings restrict entity count and wording. Deletion-based queries can misbind colors under coordination or complex modification; exact matching compounds these errors and merges or discards object, attribute, and binding evidence.
Repair. Separate Qwen3-8B Objects Parser and Color Adapter modules extract variable-length canonical entities and bound object–color pairs. With the original GRiT evidence, we check each entity and judge color after binding descriptions to its target instance. Adjacent-frame detections add confirmation for Multiple Objects.

2.1.3. Object Class

Original scoring. A sampled frame succeeds only if its GRiT labels exactly match the preset target, without normalizing case, number, or aliases; boxes and confidence scores do not enter the decision.
Mechanism. Label equality substitutes for object presence, so equivalent spellings or aliases can appear to be detection failures.
Repair. A fine-tuned Qwen3-8B Object Adapter parses the full prompt and canonicalizes case, word forms, and declared aliases before comparison with the original GRiT evidence.

2.2. Nuisance Entanglement

Nuisance Entanglement mixes relevant and irrelevant evidence.

2.2.1. Subject and Background Consistency

Original scoring. Both dimensions use whole-frame features. Subject Consistency compares DINO [16] features with the first and preceding frames; Background Consistency uses CLIP ViT-B/32 [17] with the official temporal aggregation.

Mechanism. Background, composition, and scale affect subject features; subject motion, deformation, and occlusion affect background features. Whole-frame evidence therefore couples the two targets.
Repair. Mask R-CNN [18] and MobileSAM [19] isolate the subject for pairwise cross-frame comparisons of DINO patch features. For background scoring, we independently localize foreground, extract CLIP patch tokens from its complement, and compare background regions across frames.

2.2.2. Dynamic Degree

Original scoring. At approximately 8 fps, VBench computes adjacent-frame RAFT [20] flow and averages its largest 5% magnitudes. A video is dynamic if enough pairs exceed a resolution-scaled threshold; the dataset score is the fraction classified as dynamic.
Mechanism. Local back-and-forth texture warps can inflate flow without adding coherent object or camera motion. We test stability to this nuisance while preserving motion sensitivity.
Repair. Frozen V-JEPA 2.1 ViT-B [21] encodes sixteen 384×384 RGB frames at 8 fps. A 51,393-parameter head (aligned-v1) projects tokens, attention-pools them, and applies an MLP and sigmoid. Training combines human motion preferences, native/jitter consistency, low-score still anchors, and ordered translation/reversal controls. A training-only population-mean loss aligns the scale with Origin without copying individual binary labels. Inference needs only the video, without paired references or post-hoc rescaling; scores represent relative motion, not calibrated physical intensity.

2.3. Evidence Collapse

Evidence Collapse discards discriminative information.

2.3.1. Spatial Relationship

Original scoring. GRiT boxes support spatial-relation judgments, but aggregation uses absolute horizontal or vertical displacement, $|\Delta x|$ or $|\Delta y|$.
Mechanism. Absolute displacement retains separation but loses direction. Together with lost object roles during candidate aggregation, this makes "A left of B" and "A right of B" indistinguishable under mirroring.
Repair and boundary. An independently trained Qwen3-8B Spatial Adapter maps p to ordered subject–relation–object triples $\mathcal{R} = f_\theta(p) = \{(s_i, r_i, o_i)\}_{i=1}^{N}$. Here, p is the prompt, θ denotes the adapter parameters, N is the number of extracted triples, and (s_i, r_i, o_i) is the canonical subject, relation, and object in the i-th triple. Canonicalized entities and relations align with the same GRiT detections. The full repair preserves signed subject–object displacement. A parser-only ablation keeps the unsigned backend: it improves representation and binding but cannot restore discarded direction, separating parsing from the geometric change needed for directional sensitivity.

3. EXPERIMENTS

3.1. Settings and Training

Scene tests strict synonyms on 200 videos with 16 fixed Tag2Text captions per video. Human Action tests same-class synonyms, other K400 classes, out-of-vocabulary (OOV) actions, and known-class-plus-OOV combinations on 1,200 videos with fixed UMT top-five predictions. Object Class varies case, aliases, and absent targets on 200 videos, preserving video bytes and original prompts. Subject Consistency preserves subject pixels while blurring backgrounds at varied temporal positions; Background Consistency intervenes on foreground with background fixed (200 videos).

Dynamic Degree uses 450 official VBench 1.0 MP4s from 30 previously evaluated prompts, hence not a new independent holdout. Each source has two fixed-seed 8-pixel local texture warps and an encoding control: all 1,800 inputs, including construction warnings, are scored. Bounded displacement alternates without cumulative drift, RGB noise, or brightness flicker; timing and frame count are fixed. Origin scores reuse these locked inputs.

Spatial Relationship mirrors boxes on 980 videos, with physical video mirrors and synchronized relation-word changes as controls. Multiple Objects occludes required entities with equal-area background and non-target controls (980 endpoints; 11,760 transformations). Color varies target visibility at 100/75/50/25/0%, preserving identity and other content (42 requests). Tables 1–2 give complete-pair results.

All semantic repairs use Qwen3-8B. Scene has an independent verifier; Action, Spatial, and Multiple Objects share a frozen backbone with separate LoRA [22] modules producing actions, relations, and entities. Object/Color use separate compilers. The Dynamic head uses 210 training and 60 development videos, prompt-disjoint from evaluation, continuing from the still-anchored head for 300 AdamW [23] steps. The encoder stays frozen; evaluation videos and scores never enter training.

Table 1. Invariance results (mean base → CF).

Dimension	Origin	Repair
Scene	$0.3059 \rightarrow 0.0541$	$0.6909 \rightarrow 0.6947$
Human Action	$0.8892 \rightarrow 0.0000$	$0.8892 \rightarrow 0.8892$
Object Class	$0.9911 \rightarrow 0.0000$	$0.9911 \rightarrow 0.9911$
Subject Consistency	$0.9364 \rightarrow 0.8164$	$0.9514 \rightarrow 0.9417$
Background Consistency	$0.9468 \rightarrow 0.9347$	$0.9630 \rightarrow 0.9596$
Dynamic Degree	$0.6800 \rightarrow 0.8578$	$0.6293 \rightarrow 0.6230$

Paired $n = 200/1{,}200/14/240/188$ for the first five rows; Dynamic: 450 sources, two CF seeds each (900 interventions).

Table 2. Sensitivity results (mean base → CF).

Dimension	Origin	Repair
Spatial Relationship	$0.5597 \rightarrow 0.5322$	$0.4442 \rightarrow 0.0094$
Multiple Objects	$0.4142 \rightarrow 0.0277$	$0.3983 \rightarrow 0.0178$
Color Consistency	$1.0000 \rightarrow 0.7167$	$0.9583 \rightarrow 0.1875$

Complete pairs: Spatial $n = 106$ (video mirroring), Multiple Objects $n = 505/980$, and Color $n = 3/5$. Undefined scores are excluded.

3.2. Counterfactual Results

Tables 1–2 show improved invariance and sensitivity. Under box mirroring, full Spatial repair changes $0.4442 \rightarrow 0.0000$, while Origin stays at 0.5597; Color's median endpoint score changes $1.000 \rightarrow 0.000$ on accepted bases. Dynamic Origin rises by 0.1778; Repair changes by -0.0062 (prompt-cluster bootstrap 95% CI: $[-0.0099, -0.0029]$), with mean absolute change 0.0217. This is aggregate stability: $28/900$ interventions lower Repair by more than 0.1. Human motion preference agreement is $144/171$ on originals and each jitter view, although individual pair outcomes can change.

Table 3. Dynamic Degree supplementary evaluation.

Subset (n)	Origin	Repair
Mean score: original → CF		
LAS. static (16)	$0.0000 \rightarrow 0.8750$	$0.1259 \rightarrow 0.1631$
LAS. motion (27)	$0.5926 \rightarrow 0.9815$	$0.5897 \rightarrow 0.6100$
BMC all (84)	$0.2619 \rightarrow 1.0000$	$0.2824 \rightarrow 0.2920$
Motion/static AUROC: original / CF1701 / CF2904		
LASIESTA (43)	$0.7963/0.5938/0.5127$	$0.9884/0.9630/0.9583$
BMC (64)	$0.6864/0.5000/0.5000$	$0.8712/0.9085/0.8746$

LAS.: LASIESTA. CF averages two seeds per clip; encoding controls excluded. AUROC motion/static $n = 27/16$ (LAS.), $59/5$ (BMC); 20 uncertain BMC clips enter stability means only.

Table 4. Qwen3-8B compiler ablation.

Task / reference	n	Rules	Base	LoRA
Complete JSON reference agreement (%)				
Object / seed	20	100.00	60.00	100.00
Color / seed	20	100.00	100.00	100.00
Object / silver	160	85.00	24.38	94.38
Color / silver	160	81.88	50.00	99.38
Mean original-video support score				
Object	14	0.9911	0.6339	0.9911
Color	5	0.9750	0.9750	0.9750

Base: untuned Qwen3-8B; LoRA: task-specific adapters. Reference agreement is not human-verified accuracy.

3.3. Ablations and Supplementary Evaluation

Dynamic Degree. Table 3 reruns both frozen backends on originals, encoding controls, and two jitter seeds: 43 clips from nine LASIESTA [24] recordings ($172/172$ inputs) and 84 from seven time-valid BMC [25] recordings ($336/336$), without retraining or rescaling. LASIESTA uses official object-state labels and every third frame encoded at 8 fps (native fps unverified); BMC uses verified timestamps and pre-scoring agent visual review. Construction warnings remain. BMC AUROC is exploratory: only five static clips from two recordings, with 20 uncertain clips excluded from AUROC but retained in stability means. Repair changes by more than 0.1 on $37/168$ BMC interventions. A post-training single-image diagnostic scores a still clip at 0.0074, with 8-pixel texture jitter at 0.0139, and with 8-/32-pixel translations at $0.3519/0.5549$. This controlled diagnostic does not establish broad motion validity or physical intensity calibration.

Qwen3-8B. Table 4 compares rules, untuned Qwen3-8B, and independent Object/Color LoRA adapters. Each task has 20 held-out source groups shared by 20 official seed prompts and 160 teacher-generated silver records, reported separately. JSON reference agreement includes invalid outputs; references are not human-reviewed. LoRA improves silver agreement but gives no additional mean video support over rules with fixed GRiT and Repair scoring. Support scores are not accuracies; higher scores alone do not imply better evaluation.

4. CONCLUSION

Auditing nine VBench dimensions reveals Target Substitution, Nuisance Entanglement, and Evidence Collapse. Counterfactual tests guide repairs to semantic interpretation, region selection, motion readout, and geometry. The results support targeted improvements with explicit limits: aggregate jitter stability is not physical calibration, and better parsing cannot recover direction discarded by downstream aggregation.

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